

Weitong Li

Tel: +86 189-3882-8647 | Email: weitongli2027@163.com

Education

Beijing Normal University-Hong Kong Baptist University (BNBU)

Zhuhai, China

B.S. in Artificial Intelligence | CGPA: 3.75/4.00 (Top 7%)

Sept. 2023 - June 2027 (expected)

Honor: First-Class University Scholarship (Top 6%), 2023-2024 & 2024-2025;

President's Honour Roll, BNBU, 2023-2026

Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM)

Hong Kong Baptist University

Hong Kong, China

Exchange Student

Sept. 2025 - Jan. 2026

Honor: Greater Bay Area Research & Study Scholarship (HK\$5,000)

Publication & Conference Presentations

- 【1】X. Zhu†, **W. Li†**, and J. Lin, “PRISM: Principled Reasoning for Identifying and Suppressing Model Biases at Scale,” in *PRICAI 2025: Trends in Artificial Intelligence*, Wellington, New Zealand, Lecture Notes in Computer Science, 2026, pp. 35–50, doi: 10.1007/978-981-95-7081-2_3.
- 【2】J. Lin, **W. Li**, and D. Li, “Domain-Adaptive Fine-Tuning of HI-Diff for Enhanced Microscopy Deblurring and Bacterial Flagellar Motor Localization,” in *2025 International Conference on Virtual Reality and Visualization (ICVRV)*, Bogotá, Colombia, 2025, pp. 151–156, doi: 10.1109/ICVRV67992.2025.00034.
- 【3】D. Song, **W. Li**, W. Song, W. Zheng, X. Liu, and Y. Lv, “Edge-Cloud Collaborative USV-ROV System: Active Tether Control and Multi-Agent Navigation for Marine Pastures,” in *2026 International Conference on Sensor-Cloud and Edge Computing System (SCECS)*, Xuzhou, China, 2026, pp. 20–25, doi: 10.1109/SCECS69812.2026.11566059.
- 【4】J. Lin, Y. Zeng†, **W. Li†**, and R. Lee, “G-UNet++ and Chaotic-GUNet++: Gravitational Nested U-Nets for Biomedical Image Segmentation,” *Asia Pacific Bioinformatics Conference (APBC 2026)*, Poster Track, accepted; not published in conference proceedings, 2026.

Internship

China Innovation & R&D Center, MSD (Merck) & Co., Inc.

Dec. 2025 - May 2026

AI Drug Discovery Intern

Supervised by Associate Principal Scientist Xinan Liu at MSD & Co., Inc

- Developed end-to-end AI systems for molecular generation and property prediction using BoltzGen, REINVENT4, Boltz-2, and ADMET models, integrating generative modeling, neural inference, and multi-stage evaluation.
- Designed a multi-objective Transfer-Reinforcement learning framework in REINVENT4, incorporating ADMET and Boltz-2 signals into the reward function and improving composite molecular quality scores by ~10%.
- Built a unified model benchmarking and inference pipeline across graph-based, molecular-embedding, and ADMET architectures for toxicity, physicochemical properties, and target-binding prediction.
- Engineered scalable AWS-based AI training and evaluation pipelines for reproducible multi-model experimentation, and contributed to the fine-tuning, adaptation, and evaluation of Qwen-based multimodal foundation models for biomedical applications.

(China Electronics) Digital Guangdong Chaozhou Branch

July 2024 - Aug.2024

Project Development Intern

- Coordinated with clients, prepared technical documentation, and developed multiple full-stack web interfaces by integrating front-end and back-end components, improving information presentation and supporting client-facing business workflows, with 90% client satisfaction.
- Conducted multidimensional analysis and visualization of urban and rural development data across Guangdong Province and developed an interactive data visualization platform to support comparative analysis, trend identification, and data-driven decision-making.

Research Experience

VLM and 5G Based Multi-Functional Autonomous Underwater Robots

Jan. 2026 - Sep. 2026

Supervised by Lecturer Yanyan Ji at BNBU NAVIGATOR Robotics and Innovation Lab

Published a second-author paper at SCECS 2026

Received a Gold Innovation and Entrepreneurship Scholarship (RMB 80,000)

Third Prize in the Guangdong Provincial “Challenge Cup” Extracurricular Technology Works Competition

- Integrated 5G/Wi-Fi long-range communication, power supply, and high-precision GPS positioning to enable 24/7 real-time remote control and data transmission.
- Developed a Vision-Language Model and fine-tuned YOLO-based detection models, enabling robots to detect and track fish, and supporting LLM-based interactive control and decision-making in underwater environments.
- Participated in circuit-board assembly and soldering, power distribution, sensor and communication-module integration, and embedded control development on the RK3588 platform; interfaced motor drivers and onboard sensors, implemented PID-based motion control and PWM thruster actuation, and supported real-time obstacle avoidance and system-level hardware debugging.

- Designed and fabricated a 3D-printed modular robot platform integrating embedded computing, sensing, communication, and power-control hardware; optimized layout, wiring, and waterproofing for reliable deployment.
- Developed an edge-cloud multi-agent navigation framework for coordinated USV-ROV operation, integrating perception, navigation, and tether control.
- Developed an embedded sensing and electrical integration pipeline, coordinating sensor acquisition, power regulation, communication interfaces, and actuator feedback to improve the reliability of real-time robotic operation.

Embedded G-UNet++ and Chaotic-GUNet++ for Biomedical Image Segmentation

March. 2026 – Aug.2026

Supervised by Associate Prof. Raymond Lee at BNB

Accepted as co-second author by APBC Poster track 2026 (CORE B, CCF)

Extended work in progress for submission to Connection Science (JCR Q2)

- Co-developed G-UNet++ and Chaotic-GUNet++, introducing a physics-inspired gravitational bottleneck for self-attention-free global spatial modeling and a differentiable chaotic activation mechanism for complex anatomical boundary representation.
- Evaluated the proposed architectures on ACDC cardiac MRI and MSD Spleen CT against UNet, UNet++, ACC-UNet, Swin-UNet, and G-CASCADE; Chaotic-GUNet++ achieved 85.46% mean Dice on ACDC and 90.24%
- Conducted cross-dataset transfer experiments from ACDC to MSD Spleen, where G-UNet++ achieved 94.05% Dice after target-domain fine-tuning, demonstrating strong robustness and transferability across imaging domains.

HI-Diff for Microscopy Deblurring & Bacterial Flagellar Motor Localization

June 2025 - Aug. 2025

Published a second-author paper at ICVRV 2025 (CCF)

- Adapted the HI-Diff model to microscopy datasets via domain-adaptive fine-tuning, including joint tuning, sequential tuning, and variants incorporating BBBC06 data.
- Enhanced model performance with percentile clipping normalization (p2/p98), multi-scale prior fusion (HIM module), and LoRA integration, achieving a peak PSNR of 40.31 dB.
- Applied MHAF-YOLOv10m-v2 for slice-wise detection of 3D tomographic microscopy images; optimized results using test-time augmentation and weighted box fusion methods; finalized localization via 3D-NMS for redundancy removal.

Principled Reasoning for Large-Scale Model Bias Identification & Mitigation

Mar. 2025 – June.2025

Published as co-first author at PRICAI 2025 (CORE B, CCF)

- Co-developed PRISM, a parameter-free framework for diagnosing and mitigating multi-dimensional LLM biases, integrating Dimensional Bias Identification (DBI), Targeted Example Synthesis (TES), and Discriminative Learning Enhancement (DLE) into a unified inference-time pipeline
- Designed DBI and TES, combining causal-invariance testing, constrained bias-representation decomposition, and adversarial counterfactual synthesis to isolate interpretable spurious dimensions and generate targeted debiasing signals; empirical analysis identified an optimal representation dimensionality of $k = 5$.
- Developed information-maximizing contrastive example selection for 5-shot DLE prompting, improving Llama-2 performance on UNQOVER from 23.4% to 64.5% (+41.1 pts) and HANS by 14.8 pts; ablations identified DBI as the dominant component, with a 7.3-point average drop when removed

Humanoid Robotic Skin Cancer Classification System Using Fusion Approach

Jan. 2025 - Apr. 2025

Supervised by Prof. Wentao Fan at BNB

Selected as an Outstanding AI Program Project for the BNB Faculty of Science and Technology 2025 Poster Exhibition

- Developed a multimodal skin-disease diagnosis system and integrated it into a self-designed, 3D-printed humanoid robot for preliminary skin-disease assessment and human-robot interaction.
- Created a cascaded CNN-LightGBM model that combines image-based predictions with structured clinical features, achieving a macro-F1 of approximately 0.94 and 100% recall on the evaluation set.
- Completed end-to-end system integration covering image acquisition, multimodal feature fusion, model inference, and robotic deployment, demonstrating the practical application of AI models in an embodied healthcare scenario.

PROJECT EXPERIENCE

Multi-Agent Medical Consultation System Based on OpenClaw

Feb. 2026 - Apr. 2026

- Developed a multi-agent medical consultation system integrating diagnosis, appointment scheduling, and personalized recommendations using LangChain Agents and RAG.
- Designed an end-to-end consultation workflow covering symptom collection, diagnostic reasoning, information retrieval, and recommendation generation, achieving approximately 80% diagnostic accuracy in evaluation.

Multi-Agent Relationship Prediction

Feb. 2026 - Apr. 2026

- Developed a multi-agent interaction platform supporting one-to-one, one-to-many, and many-to-many relationship scenarios, enabling dynamic simulation and prediction of agent interactions.
- Designed graph-based relationship modeling and dynamic affinity-update mechanisms to represent evolving interpersonal states and predict subsequent interaction behaviors.
- Integrated the ReAct framework for agent reasoning and action planning, enabling agents to iteratively interpret contextual information, update relationship states, and generate adaptive interaction strategies.

SKILL

Python, Java, NLP, Machine Learning, Computer Vision, Embedded AI, RK3588, PID/PWM, Circuit Assembly & Soldering